4.

DNN AFFINE (DENSE) $A^{(L)} = A^{(L-1)} \cdot W^{(L),T} + \vec{b}^{(L),T}$ ReLU if $x<0$ then $x=0$

SOFTMAX $S_i = \frac{e^{z_i}}{\sum e^{z_k}}$ $\frac{e^x}{e^x e^x} = \frac{e^x}{2e^x} = \frac{1}{2}$ # classes = size last layer

ACTIVATION (TRANSFER) FUNCTIONS: ReLU, S, sin, cubic (apply to all elements)

LAYER SHAPE $SH(W^{(L),T}) = (n_in, n_out)$ $SH(b^{(L)}) = (n_out)$

FREE PARAMETERS DENSE: $(n_out \cdot n_in) + n_out$ filters ReLU, SOFTMAX, MAX POOL = 0

CNN Conv2D param = nº filter ($fH \cdot fW \cdot C_in$) + nº. filters C_out = nº of filters

too!

LAYER DIMENSIONS: $H_out = floor((H_in - filter)/stride) + 1$

MAX POOL $H_{out} = H_{in}/kernel$ $C_{out} = C_{in}$ CONV2D FLATTEN $H = W \times H \times C$

THEORY DROPOUT zero fraction p during training, prevents overfitting.

Leaky ReLU: αx for $x<0$, fix dying gradients. ADAM MOMENTUM (β_1) + ADAPTIVE (β_2) layer, fast convergence

SEQUENTIAL: linear chain

FUNCTIONAL: skip connections, arbitrary topology

model(): tensor, small

• predict(): numpy, large

• evaluate(): loss + metrics

conv layers, fix vanishing gradients

PADDING = "SAME" output same H, W (in skip connections) SKIP CONN.: add input around

Pitfalls: normalize to [0,1], class imbalance → oversample

CONFUSION MATRIX 5.

PROBABILITIES

elementary: $p(A) = \frac{\#(A)\ C_{ij}}{N}$ $A \Rightarrow \hat{y}=k$ $B \Rightarrow \hat{t}=j$

JOINT: $p(A,B) = \frac{\#(A \text{ and } B)}{N}$ $A \text{ and } B \Rightarrow C_{ij}$

CONDITIONAL: $p(A|B) = \frac{\#(A \text{ and } B)}{\#B}$ $p(B|A) = \#(A \text{ and } B) / \#A$

incorrect classif: $(\hat{y} \neq \hat{t})$ restrict to class $(\hat{t} = class)$ restrict classif output $(\hat{y} = out)$

$\hat{y}$ = row TRUE, $\hat{t}$ = col PREDICTED

↳ $\frac{\Sigma \text{ incorrect classif in class}}{\Sigma \text{ all } \hat{y}}$ output $= \frac{\Sigma \text{ incorrect predictions in } \hat{t}}{\Sigma \text{ all } \hat{t}}$

N = nº of pairs (simple probab.) OR in CM, sum of all element values!

$p(\hat{y}=1 | \hat{t}=3)$ → chance of predicting class 1 if class 3 is true.

$\hat{y}$ = row, true, classifier

$\hat{t}$ = col, predicted, sample

$CM = \begin{pmatrix} tn & fp \\ fn & tp \end{pmatrix}$ (2,2)

p: n ∈ given

$p_p = p/N$

$p_n = n/N$

$N = p + n$ ∑ row if pred

BINARY CLASSIFICATION

$tpr = tp/(tp+fn) = p(\hat{y}=2, \hat{t}=2) = 1 - fnr$ = sensitivity = recall

$tnr = tn/(tn+fp) = p(\hat{y}=1, \hat{t}=1) = 1 - fpr$ = speaficity

$fpr = fp/(tn+fp) = p(\hat{y}=2, \hat{t}=1) = 1 - tnr$

$fnr = fn/(tp+fn) = p(\hat{y}=1, \hat{t}=2) = 1 - tpr$

PRECISION $= tp/(tp+fp)$ ERROR then ACCURACY

$E = fnr \cdot pp + fpr \cdot pn$

$Acc = 1 - E$

LOSS FUNCTIONS

CROSS ENTROPY

$\mathcal{L}^{CE}(Y|T) = -\frac{1}{N} \sum_i \sum_j \log(Y_{i,k}) \cdot T_{i,k}$

MEAN SQUARE ERROR

$\mathcal{L}^{MSE}(Y|T) = \frac{1}{N} \sum_{i,k} (Y_{i,k} - T_{i,k})^2$

RMSE

RMSE $\mathcal{L} = \sqrt{\sum_i (Y_i - T_i)^2}$

GRADIENT $PYL = (Y-T)/L$

CLASSIF. ERROR $\mathcal{L}^{CC}(Y|T) = \frac{1}{N} \sum_{i=1}^{N} \begin{cases} 1 \rightarrow \text{for each row} \\ 0 \text{ w/ wrong classif.} \end{cases}$